On the Duality Between Quantized Time and States in Dynamic Simulation

Liya Huang and Georgios Tzounas, *Member, IEEE*

***Abstract*—This letter introduces a formal duality between discrete-time and quantized-state numerical methods. We interpret quantized state system (QSS) methods as integration schemes applied to a dual form of the system model, where time is seen as a state-dependent variable. This perspective enables the definition of novel quantized state system-based schemes inspired by classical time-integration techniques. As a proof of concept, we illustrate the idea by introducing a QSS-based Adams–Bashforth method applied to a test equation. We then move to demonstrate how the proposed approach can achieve notable performance improvements in realistic power system simulations.**

***Index Terms*—Time-domain integration, quantized state system, quantum control, duality.**

I. Introduction

THE numerical simulation of power system dynamics inherently requires discretization. Classical Runge–Kutta and multistep methods are based on time discretization, where the system dynamics are approximated at a sequence of discrete time points. An alternative paradigm is offered by the family of quantized state system (QSS) methods [1], [2], [3], where the core idea is to quantize the state variables instead of discretizing time. These methods can offer computational advantages in problems with localized activity, where some states require more frequent updates than others. This letter introduces a new perspective that exploits classical integration techniques to guide the design of novel and useful quantized state system-based schemes.

The family of QSS methods was systematically introduced by Kofman within the discrete event system specification (DEVS) framework [1]. The development began with the first-order quantized state system (QSS1) method, and was subsequently extended to higher-order variants designed to improve accuracy [2], [3]. A limitation of these methods is the need to estimate higher-order state derivatives during simulation, which increases computational cost. Linearly-implicit variants (LIQSS) that rely on local linearization and explicit predictors were later proposed to enhance numerical performance for stiff systems; see, e.g., [4]. The application of QSS methods to power systems has so far been limited. Existing studies have primarily considered the simulation of single converters and synchronous machines connected to an infinite bus [4], [5]. A key reason behind this limited application is the asynchronous nature of QSS methods, where, in principle, each state variable updates at its own rate. This structure complicates their use in large, coupled systems that require coordinating state updates.

In this letter, we revisit quantized state system methods through a new lens by drawing analogies with classical time-integration schemes, and derive a dual formulation in which time is treated as a state-dependent variable. This dual view allows us to improve the accuracy of quantization event timing and to define a novel QSS-based variable-step mechanism compatible with standard power system solvers.

II. Duality of State and Time Discretization

A. Preliminaries

The central idea behind quantized state system is to approximate the solution of differential equations by discretizing the states rather than time. The simplest and most intuitive variant is the first-order scheme, QSS1. Consider the scalar equation:

$$x'(t) = f(x(t)) \quad x \in \mathbb{R} \tag{1}$$

QSS1 tracks $x(t)$ and updates a piecewise-constant signal $q(t)$ whenever $x(t)$ changes by a fixed amount Δq, called the *quantum*. The signal $q(t)$ acts as a quantized approximation of $x(t)$, initialized as $q(0) = x(0)$, and defined by [1]:

$$q(t) = \begin{cases} x(t) & \text{if } |x(t) - q(t^-)| \geq \Delta q \\ q(t^-) & \text{otherwise} \end{cases}$$

At any time t, the state derivative is evaluated using $q(t)$. That is, (1) is approximated as:

$$x'(t) = f(q(t)) \tag{2}$$

Since $q(t)$ changes only when $x(t)$ deviates from it by more than Δq, the right-hand side remains constant in-between, i.e., between *quantization events (QE)*. Thus, in QSS1, $x(t)$ is piecewise linear between successive updates of $q(t)$. Let t_k denote the time of the k-th QE. At this time, we have that $q(t_k) = x(t_k)$. Then, for $t \in [t_k, t_{k+1})$, the state evolves as:

$$x(t) = x(t_k) + (t - t_k) f(x(t_k)) \quad t \in [t_k, t_{k+1}) \tag{3}$$

The next QE occurs at $t = t_{k+1}$, when, by definition, the following condition is satisfied:

$$|x(t) - x(t_k)| = \Delta q \tag{4}$$

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The authors are with the School of Electrical and Electronic Engineering, University College Dublin, D04V1W8 Dublin, Ireland (e-mail: georgios.tzounas@ucd.ie).

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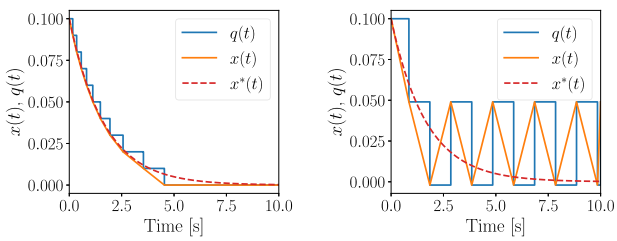
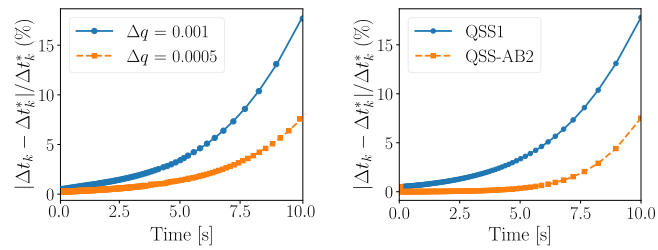


(a) $\Delta q = 0.01$. (b) $\Delta q = 0.051$.

Fig. 1. Test equation $x' = -0.6x$: QSS1 solution.



(a) QSS1. (b) $\Delta q = 0.001$.

Fig. 2. Test equation $x' = -0.6x$: Event quantization timing errors.

Substituting (3) in (4) at $t = t_{k+1}$, we get $\Delta t_k |f(x(t_k))| = \Delta q$, where $\Delta t_k = t_{k+1} - t_k$. The integration time step is:

$$t_{k+1} = t_k + \Delta q / |f(x(t_k))| \tag{5}$$

and represents the estimated time at which the next *QE* occurs.

B. Duality and QE Timing

Equation (5) reveals a formal analogy, specifically, it is equivalent to forward Euler method applied to the following *dual system* where time evolves as a state-dependent variable:

$$t'(x) = \phi(t(x)) \tag{6}$$

with $\phi = 1/|f|$. That is, $t_{k+1} = t_k + \Delta q\ \phi(t_k)$. This dual perspective allows us to interpret quantized state system methods as discrete approximations of (6). The following implications are relevant: (i) if $x' = 0$, then $t' \to \infty$. In other words, quantized state system jumps directly to the final simulation time if the system has reached steady state, skipping unnecessary computations. (ii) if $x' \to \infty$, then $t' \to 0$. In other words, quantized state system becomes inefficient in the presence of high-frequency noise or instability. A trade-off between efficiency and temporal accuracy can be achieved by enforcing a suitable minimum time increment.

The duality above offers valuable insight into the numerical behavior of quantized state system methods. In particular, since (5) mirrors forward Euler applied to (6), it follows that QSS1 inherits its numerical instability. In this case, however, the instability does not cause unbounded growth of the state error, but rather *destabilizes the QE timing*: when Δq is too large, computed event times may deviate significantly from the point at which the state actually crosses the quantization thresholds. This motivates the design of novel higher-order methods. In this letter, we introduce a second-order Adams–Bashforth variant:

$$t_{k+1} = t_k + \Delta q\ [1.5\phi(t_k) - 0.5\phi(t_{k-1})] \tag{7}$$

These two enhancements are independent and can be conveniently combined. (7) improves the derivative approximation and provides a more accurate estimation of QE timing. Thus, it is conceptually different from higher-order variants which focus on state reconstruction by making $q(t)$ piecewise linear (QSS2) or parabolic (QSS3). For simplicity, in this letter we retain the piecewise-constant (QSS1) structure of $q(t)$. In addition, the idea is not limited to AB2; other schemes can also be formed within the same dual framework.

As an illustrative example, Fig. 1 shows the numerical solution of the test equation $x'(t) = -0.6x(t)$, with $x(0) = 0.1$, obtained using QSS1. For reference, the exact analytical solution $x^*(t)$ is also included. As expected, increasing Δq leads to larger numerical errors. For sufficiently large Δq (in this case for $\Delta q > 0.05$), QSS1 is no longer able to track the exponential state decay. The effect of the quantum size Δq on the QE timing accuracy of QSS1 is illustrated in Fig. 2(a). Since the analytical solution $x^*(t)$ is available for the test equation, the exact time interval Δt_k^* at which the state reaches the next QE threshold is computed directly and used as benchmark for comparison. The relative error $|\Delta t_k - \Delta t_k^*|/\Delta t_k^*$ quantifies the deviation between the computed time step Δt_k and the exact interval Δt_k^*. As shown in Fig. 2(b), QSS-AB2 significantly reduces this error, providing improved event timing accuracy under the same Δq.

III. QSS-Based Time Step Control

In the preceding analysis we used a scalar equation to illustrate the key idea of viewing quantized state system as a time-discretization applied to a dual system, where the roles of time and state are interchanged. We now generalize this perspective to systems of equations, focusing on dynamical models that describe the behavior of power systems. These models can be expressed in the form of explicit differential algebraic equations (DAEs):

$$\begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \mathbf{0}_m \end{bmatrix} \begin{bmatrix} \boldsymbol{x}'(t) \\ \boldsymbol{y}'(t) \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{y}(t)) \\ \boldsymbol{g}(\boldsymbol{x}(t), \boldsymbol{y}(t)) \end{bmatrix} \tag{8}$$

where $\boldsymbol{x}(t) \in \mathbb{R}^n$ and $\boldsymbol{y}(t) \in \mathbb{R}^m$ are the state and algebraic variables, respectively. The functions $\boldsymbol{f}$ and $\boldsymbol{g}$ define the differential and algebraic equations, while $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{0}_{n\times m}$ the $n \times m$ zero matrix. We associate a local dual system to each differential equation, as follows:

$$t_i'(\boldsymbol{x}, \boldsymbol{y}) = \phi_i(t_i(\boldsymbol{x}, \boldsymbol{y})) \tag{9}$$

where $\phi_i = 1/|f_i(\boldsymbol{x}, \boldsymbol{y})|$ for $i \in \{1, \ldots, n\}$. By applying (7), we obtain n candidate local time steps:

$$\Delta t_{k,i} = \Delta q[1.5\phi_i(t_{k,i}) - 0.5\phi_i(t_{k-1,i})] \tag{10}$$

Implementing such asynchronous system updates raises two issues. First, algebraic equations do not have finite, nonzero time constants, preventing the application of QSS-based stepping to the full DAEs. Second, each QE may affect multiple variables

whose dynamics are coupled, but the causal ordering of their updates is not uniquely defined and thus depends on heuristics. In this letter we address these issues by imposing a global time step for the entire system. In particular, we compute $\Delta t_{k,i}$ for each equation from (10) and select:

$$\Delta t_k = \min\{\Delta t_{k,i}\} \quad i \in \{1, \ldots, n\} \tag{11}$$

This allows combining novel QSS-based schemes like (10) with standard solvers such as the Trapezoidal Method (TM).

We note that in this formulation, QEs are not enforced; instead, Δq acts as a parameter that governs time step-size control. This ensures that no state overshoots its quantization threshold, but can lead to conservative results, as all equations are synchronized with the most sensitive one. To reduce conservativeness and improve efficiency, we allow Δq to vary dynamically during the simulation, thereby indirectly controlling the timing of QEs. Specifically, we propose updating the Δq using a PI control, directly inspired by classical step-size adaptation strategies based on local error estimates [6], [7]:

$$\Delta q_{k+1} = (\mathrm{tol}/|\sigma_k|)^{\alpha} \left(|\sigma_{k-1}|/|\sigma_k|\right)^{\beta} \Delta q_k \tag{12}$$

where σ_k is the estimated event timing error at t_k; tol is a prescribed error tolerance; and α, β are control parameters.

The combination of (10)–(12), that is, time-step selection based on duality-inspired QSS schemes, standard implicit solvers, and adaptive quantum size control, defines a novel framework that we further explore in the next section.

IV. Case Study

This section presents simulation results based on a dynamic model of the all-island Irish transmission system (AIITS) [8]. The model consists of 1503 buses, 1,851 lines and transformers, 22 synchronous machines, 169 wind power plants and 245 loads. Simulations are carried out with Dome [9], on a computer equipped with an Intel Xeon E3-1245 v5 processor, 16 GB of RAM, and a 64-bit Linux OS. Three disturbances are considered: 1) a three-phase fault at bus 1238 occurring at $t = 1$ s and cleared by tripping a line connected to the faulted bus after 80 ms; 2) a loss of 142 MW load connected to buses 91-102; 3) outage of 167 MW synchronous generation connected at bus 1236. All tests are carried out using TM as the underlying integration method, considering a simulation duration of 20 s. For each disturbance, the following configurations are evaluated: i) fixed time step; ii) QSS1-Sync, where $\Delta t = \min\{\Delta q \phi_i(t_{k,i})\}$, $i \in \{1, \ldots, n\}$, with constant Δq; iii) QSS2-Sync, where $\Delta t = \min\{\Delta q \phi'_i(t_{k,i})\}$, $\phi' = 1/|f'|$, with constant Δq; iv) LIQSS-Sync, where $\Delta t = \min\{\Delta q \phi_i(t_{k+1,i})\}$, with constant Δq; v) QSS-AB2, where Δt_k is defined through (10), (11), with constant Δq; vi) QSS-AB2-Ad, where Δt_k is defined through (10), (11), with adaptive Δq updated via the PI control rule (12). Extensive simulations are carried out for different scenarios to test the robustness of the parameters α and β, with the tolerance fixed at $\mathrm{tol} = 0.2 \times 10^{-1}$ to satisfy the accuracy requirement. As shown in Fig. 3, the simulation runtime under the three-phase fault over the range $\alpha \in [0.2, 0.7]$ and $\beta \in [0, 0.4]$ remains below 5.25 s. In the following, we chose $\alpha = 0.5$, $\beta = 0$. For

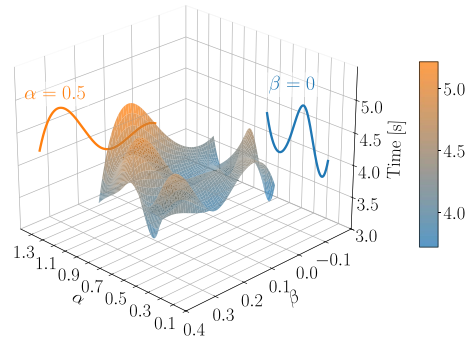


Fig. 3. Three-phase fault: Runtime for $\alpha \in [0.2, 0.7]$, $\beta \in [0, 0.4]$.

TABLE I
Performance Comparison Under Three Scenarios

Time step	Quantum	Time [s]	Avg. error [$\times 10^{-5}$]
Three-phase fault			
Fixed [$\Delta t = 0.001$ s]	-	32.2	-
Fixed [$\Delta t = 0.01$ s]	-	5.8	42.9
QSS1-Sync	Fixed [$\Delta q = 0.01$]	9.2	42.8
QSS1-Sync	Fixed [$\Delta q = 0.24$]	5.6	71.9
QSS2-Sync	Fixed [$\Delta q = 0.01$]	167.9	22.3
LIQSS-Sync	Fixed [$\Delta q = 0.01$]	194.7	14.7
QSS-AB2	Fixed [$\Delta q = 0.24$]	4.6	7.26
QSS-AB2-Ad	PI control	4.1	0.14
Load loss			
Fixed [$\Delta t = 0.001$ s]	-	47.5	-
Fixed [$\Delta t = 0.01$ s]	-	6.3	153.5
QSS1-Sync	Fixed [$\Delta q = 0.01$]	10.0	153.8
QSS1-Sync	Fixed [$\Delta q = 0.24$]	5.6	189.8
QSS2-Sync	Fixed [$\Delta q = 0.01$]	151.4	37.1
LIQSS-Sync	Fixed [$\Delta q = 0.01$]	185.7	49.7
QSS-AB2	Fixed [$\Delta q = 0.24$]	5.1	8.34
QSS-AB2-Ad	PI control	4.3	0.06
Synchronous machine outage			
Fixed [$\Delta t = 0.001$ s]	-	29.2	-
Fixed [$\Delta t = 0.01$ s]	-	4.8	94.5
QSS1-Sync	Fixed [$\Delta q = 0.01$]	9.8	94.7
QSS1-Sync	Fixed [$\Delta q = 0.24$]	4.0	121.5
QSS2-Sync	Fixed [$\Delta q = 0.01$]	125.4	24.7
LIQSS-Sync	Fixed [$\Delta q = 0.01$]	191.6	124.7
QSS-AB2	Fixed [$\Delta q = 0.24$]	3.8	1.7
QSS-AB2-Ad	PI control	2.7	0.9

QSS-AB2-Ad, the initial Δq is set to 0.2 and a maximum value of 4 is imposed to avoid excessive time steps. The error σ_k is taken as the difference between Δt_k computed with Δq_k and $\Delta q_k/2$, similar to embedded error estimates used in variable-step methods.

Table I reports the performance of the examined configurations under the examined disturbances. In both cases, the numerical error is measured for the rotor speed of a representative machine (at bus 1237), using TM with fixed step size 0.001 s as the reference. Increasing the step to 0.01 s yields faster simulations and a moderate error increase. Using QSS1-Sync with $\Delta q = 0.01$ exhibits similar accuracy to TM with $\Delta t = 0.01$ s but results in higher runtime. The results show that the simulation runtime is significantly longer, while the accuracy improvement is marginal. The increased computational time arises from the need to evaluate higher-order derivatives, which substantially increase the computational burden. The limited

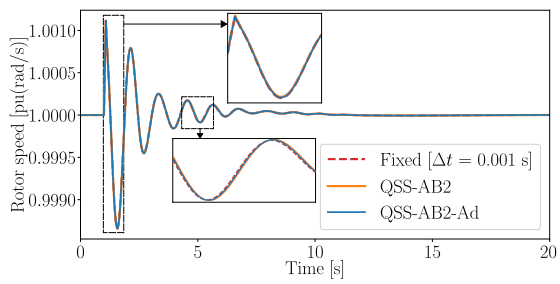


Fig. 4. Three-phase fault: Rotor speed of machine at bus 1237.

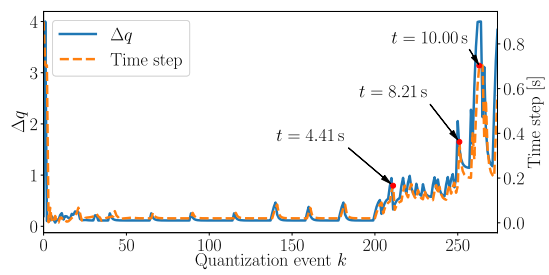


Fig. 5. Three-phase fault: Variations of Δq and Δt, QSS-AB2-Ad.

accuracy improvement occurs because higher-order methods depend on the estimation of second-order derivatives during the simulation process, introducing additional numerical errors. QSS2-Sync with $\Delta q = 0.01$ leads to the longest runtime but yields only marginal accuracy gains due to the costly estimation of higher-order derivatives for $x(t)$ and associated numerical errors. Also, the prediction step in LIQSS-Sync deteriorates the simulation efficiency. Increasing Δq to 0.24 improves runtime at the expense of accuracy. In contrast, QSS-AB2 with the same $\Delta q = 0.24$ achieves lower runtime and much lower error compared to TM with $\Delta t = 0.01$ s. Finally, QSS-AB2-Ad dynamically adjusts Δq to further improve both accuracy and runtime.

The rotor speed of the machine at bus 1237 following the fault is shown in Fig. 4. Both QSS-AB2 and QSS-AB2-Ad accurately track the reference solution, consistent with the errors reported in Table I. For the same disturbance, Fig. 5 further illustrates the behavior of QSS-AB2-Ad, showing how Δq and Δt evolve over the course of QEs during the simulation. In particular, both remain small during the transient and increase significantly as the system approaches steady state.

V. Conclusion

This letter introduces a novel perspective on quantized-state simulation by formalizing a duality between time and state discretization. To illustrate this perspective, we define a QSS-based, second-order Adams–Bashforth method and describe how it can be combined with standard implicit power system solvers to achieve significant speedups while maintaining accuracy. Further performance gains through adaptive quantum control are also discussed. The proposed approach opens directions for a new family of numerical methods that integrate time discretization and state quantization.

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